CPL rotation and pin 1 of polarized parts (tools/cpl_pin1.py)

來源：v2_r7f/fab_r7f/CPL_polarity_pin1_r7f.md（原文英文）

CPL rotation and pin 1 of every polarized part (r7e, REVIEW_r7d M1).

For each polarized part on the CPL (ICs, connectors, the crystal oscillator, polarized capacitors; plain R/C/L/FB/TP are not polarized):

- pin 1 (A1 for the BGA, A1B12 for USB-C) from design.json, as an offset from the part centre in the JLC top preview (x right, y down, = KiCad screen) and in the JLC bottom preview (x mirrored);
- pin 1 in the footprint's own frame (rotation and bottom mirror undone; the convention is checked: every part of one footprint gives the same vector whatever its angle and side);
- the JLC angle offset (CPL = KiCad + offset) from data/jlc_offsets.json. Only parts a person verified in JLC's preview have one; for every other part the offset is UNKNOWN (not 0) and the row says CHECK IN PREVIEW. FAIL when: a verified part's CPL angle is not KiCad + offset; its pin 1 (footprint frame) is not where it was when verified (footprint changed); a polarized part is missing from the CPL; a polarized capacitor has pin 1 on GND or pin 2 off GND, or its + pin does not reach a positive supply through at most one resistor (C69: + on 5V_DAMP, R68 to 5V, REVIEW_r7d M1).

Ref	LCSC	Land	Side	KiCad deg	CPL deg	JLC offset	Pin 1	Net	Pin 1 in JLC top view (mm)	Pin 1 in JLC bottom view	Status
C69	C347552	CP_7343	Bottom	0	0	UNKNOWN	1	5V_DAMP	right (+3.10, +0.00)	left (-3.10, +0.00)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
D1	C48260	X1SON-2_0.6x1	Top	0	0	UNKNOWN	1	HDMI_CEC	left (-0.35, +0.00)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J2	C720616	HDMI-001S	Bottom	180	180	UNKNOWN	1	D2_P	down-right (+4.50, +3.36)	down-left (-4.50, +3.36)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J3	C427493	USB-C_HX_16P	Top	180	180	UNKNOWN	A1B12	GND	down-right (+3.20, +2.55)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J4	C160405	JST_SM06B	Bottom	0	0	UNKNOWN	1	PWR_C	up-right (+2.50, -2.00)	up-left (-2.50, -2.00)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J5	C427493	USB-C_HX_16P	Top	90	90	UNKNOWN	A1B12	GND	down-left (-2.55, +3.20)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J6	C160407	JST_SM08B	Top	90	90	UNKNOWN	1	3V3	down-left (-2.00, +3.50)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
J7	C585350	microSD_Molex_104031-0811	Bottom	270	90	180	1	SD_D2_C	up-left (-5.45, -3.10)	up-right (+5.45, -3.10)	OK (offset verified in JLC preview 2026-11-07)

Ref	LCSC	Land	Side	KiCad deg	CPL deg	JLC offset	Pin 1	Net	Pin 1 in JLC top view (mm)	Pin 1 in JLC bottom view	Status
J8	C88373	U.FL-R-SMT-1	Bottom	0	0	UNKNOWN	1	LNA_OUT	down (+0.00, +1.52)	down (-0.00, +1.52)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
Q1	C15127	SOT-23	Bottom	0	270	270	1	SD_PWR_N	down-right (+0.95, +0.94)	down-left (-0.95, +0.94)	OK (offset verified in JLC preview 2026-11-07)
U10	C92243	SSOP-4_1.27	Bottom	0	270	270	1	PWR_LED	up-right (+3.17, -0.63)	up-left (-3.17, -0.63)	OK (offset verified in JLC preview 2026-11-07)
U11	C92243	SSOP-4_1.27	Bottom	0	270	270	1	RST_LED	up-right (+3.17, -0.63)	up-left (-3.17, -0.63)	OK (offset verified in JLC preview 2026-11-07)
U12	C92243	SSOP-4_1.27	Bottom	0	270	270	1	LED_IN	up-right (+3.18, -0.64)	up-left (-3.18, -0.64)	OK (offset verified in JLC preview 2026-11-07)
U14	C138714	USON-10	Top	0	0	UNKNOWN	1	USB2_DP	down-left (-1.00, +0.41)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U15	C78988	SOT-23-5	Top	90	0	UNKNOWN	1	3V3	down-right (+1.14, +0.95)	-	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U16	C2155674	WSON-8_2x2_EP	Bottom	90	90	UNKNOWN	1	EF_DVDT	down-right (+0.75, +0.95)	down-left (-0.75, +0.95)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U2	C78988	SOT-23-5	Bottom	0	270	270	1	5V	down-right (+0.95, +1.14)	down-left (-0.95, +1.14)	OK (offset verified in JLC preview 2026-11-07)
U3	C138714	USON-10	Bottom	90	90	UNKNOWN	1	USB_DN	down-left (-0.41, +1.00)	down-right (+0.41, +1.00)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U4	C882825	SOT-23-5	Bottom	0	270	270	1	3V3	down-right (+0.95, +1.14)	down-left (-0.95, +1.14)	OK (offset verified in JLC preview 2026-11-07)
U6	C201698	MSOP-8	Bottom	90	90	UNKNOWN	1	3V3	down-left (-2.11, +0.98)	down-right (+2.11, +0.98)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U7	C3008606	BGA-64_6x6	Bottom	180	180	UNKNOWN	A1	TC_REXT	down-left (-2.27, +2.27)	down-right (+2.27, +2.27)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say

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U8	C138714	USON-10	Bottom	180	180	UNKNOWN	1	D1_N	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
U9	C138714	USON-10	Bottom	180	180	UNKNOWN	1	CK_N	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
Y1	C725995	OSC3225	Bottom	0	0	UNKNOWN	1	OSC_VDD	down-right (+1.05, +0.83)	down-left (-1.05, +0.83)	CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say
C69	C347552	CP_7343					+/-	+5V_DAMP, -GND			polarity OK

RESULT PASS: 24 polarized parts, 7 with a verified JLC offset, 17 UNKNOWN (check in JLC preview)